

MRITYUNJAY JHA

+91-8287299817 | mrityunjay.jha2005@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

EDUCATION

Delhi Technological University (Formerly Delhi College of Engineering)

Bachelor of Technology in Computer Science Engineering — CGPA: 7.94

Delhi, India

Aug 2023 – May 2027

Govt. Boys Sr. Sec. School

Delhi, India

Senior Secondary (Class XII) — Percentage: 86.4%

2021 – 2022

Ring Midways Sr. Sec. Public School

Delhi, India

Secondary (Class X) — Percentage: 94.8%

2019 – 2020

WORK EXPERIENCE

Software Engineer Intern

Nov 2025 – Present

QuickIntell

Delhi, India

- Enforced granular role-based access control across 15+ application modules using CASL and Prisma, ensuring secure invite lifecycles and strict compliance for sensitive PHI records.
- Orchestrated an idempotent, event-driven billing pipeline via Stripe Webhooks to manage multi-tenant subscriptions and usage-based overages, effectively preventing critical revenue leakage errors.
- Devised an asynchronous background processing engine, implementing fault-tolerant batching to handle heavy loads without service degradation while respecting strict external API limits.
- Operationalized an AI-driven RAG pipeline on AWS EC2, processing 3,000+ complex rows via persistent tmux sessions and indexing 50,000+ vector entries in Pinecone for precise automated mapping.

Web Development Intern

Mar 2025 – May 2025

University School of Innovation and Entrepreneurship (USIP), DTU

Delhi, India

- Authored comprehensive technical documentation for internal web platforms, ensuring code maintainability and clearer API specifications.
- Implemented frontend performance optimizations, including search debouncing and customized form validation, to improve UI responsiveness.
- Collaborated on bug fixes and minor feature updates while gaining hands-on proficiency in full-stack web development standards.

PROJECTS

NIRVANA | React.js, TypeScript, Tailwind, Node.js, Express, Hono.js, PostgreSQL, Prisma, MongoDB

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- Developed a full-stack travel blogging platform with secure authentication, CRUD workflows, and real-time one-to-one chat functionality.
- Designed a split-route backend separating WebSocket chat from CRUD APIs, reducing latency by 30% under peak load.
- Applied polyglot persistence using PostgreSQL for structured blog/auth data and MongoDB for high-throughput chat message storage.
- Integrated Prisma Accelerate to enable global connection pooling for PostgreSQL in serverless Workers, preventing database connection exhaustion under high concurrency.

ALASKA | React.js, TypeScript, C++, Cloudflare Workers, PostgreSQL, Prisma, Socket.io, Mapbox

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- Developed a route-based social matching platform that connects users based on overlapping metro commute paths, enabling personalized discovery and interaction.
- Implemented an interactive metro route builder using Mapbox GL and Dijkstra's algorithm, allowing users to generate and store customized shortest-path journeys.
- Built a C++ route-overlap matching engine using Suffix Automata, reducing ordered subpath matching complexity from $O(N \cdot L^2)$ to $O(N \cdot L)$ across stored user routes.
- Proposed an optimization using Huffman encoding to compress frequently visited station identifiers, improving matching efficiency across large route datasets.

THREE-JS-LIBRARY | React.js, Three.js, React Three Fiber, Rapier, GLSL, DracoLoader, Blender

[Link](#) | [GitHub](#)

- Developed a Three.js-based 3D rendering library featuring 9+ interactive models with smooth real-time controls and high-performance visualization.
- Reduced 3D asset load times by 50% using Draco mesh compression and optimized preprocessing workflows.
- Built GPU-accelerated particle simulations with custom GLSL shaders, sustaining 60+ FPS on mid-tier devices.
- Improved rendering throughput by 60% through shader-level optimizations and efficient scene management techniques.

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, TypeScript, HTML

Web Development: CSS, React.js, Next.js, Node.js, Express.js, REST APIs, WebSockets

Databases & ORM: PostgreSQL, MongoDB, MySQL, Prisma, Pinecone

Deployment & DevOps: Git, GitHub, Docker, Vercel, Cloudflare Workers, Linux

Graphics & 3D: Three.js, React Three Fiber, Rapier, GLSL, Shaders, DracoLoader, Blender

Relevant Coursework: Operating Systems, Computer Networks, Database Management Systems, Object-Oriented Programming

ACHIEVEMENTS

Knight on LeetCode with 500+ problems solved, earned 9 badges, and achieved a peak rating of 1860.

Pupil on Codeforces, participated in 10+ contests, and attained a peak rating of 1370.